

$$13 \quad g_s = \frac{GM}{R^2}$$

$$\text{At } 2R \text{ above the surface, } g_s' = \frac{GM}{[3R]^2} = \frac{GM}{9R^2} = \frac{1}{9} g_s$$

Answer: A

$$14 \quad \frac{GM_{\text{sun}} M_{\text{satellite}}}{r^2} - \frac{GM_{\text{earth}} M_{\text{satellite}}}{R^2} = M \left(\frac{4\pi^2}{T^2} \right) R$$